ON-BOARD DIAGNOSTIC SYSTEM [ELECTRICAL SUPPLY UNIT (ESU)]

id094000050200

Outline

- The on-board diagnostic function consists of the following functions:
  - — A malfunction detection function, which detects overall malfunctions in the electrical supply unit (ESU)-related parts.
  - — A memory function, which stores detected DTCs.
  - — A display function, which indicates malfunction locations and status via DTC output.
- Using the Mazda Modular Diagnostic System (M-MDS), DTCs can be read out and deleted.
- The on-board diagnostic function for the electrical supply unit (ESU) is performed via gateway communication of the LIN communication circuit between it and the body control module (BCM).

Malfunction detection function

- Detects malfunctions in input/output signals.
- If a malfunction occurs, the electrical supply unit (ESU) records the malfunction as a DTC. A recorded DTC can be read by the Mazda Modular Diagnostic System (M-MDS).

DTC table

×: Applicable
—: Not applicable

DTC No.	Warning/ indicator light	Description	Fail-safe	Drive cycle	Self test type*1	Memory function
B1098:11	—	Parking light (LH) or taillight (RH) circuit malfunction	—	—	C	×
B1099:11	—	Parking light (RH), taillight (LH), or license plate light (LH)/(RH) circuit malfunction	—	—	C	×
B111E:12	—	**4SD** • Trunk compartment light circuit malfunction **5HB** • Cargo compartment light circuit malfunction	—	—	C	×
B1142:13	—	Ignition switch status mismatch	—	—	C	×
B1207:09*2	—	Error signal received from SAS control module	—	—	C	×
B1207:86*2	—	Error signal received from SAS control module	—	—	C	×
B124A:11*3	—	Running light circuit malfunction	—	—	C	×
B1277:11	—	Back-up light circuit malfunction	—	—	C	×
B1313:12	—	Front map light circuit malfunction	—	—	C	×
B143A:00*2	—	Collision detection unlock function operation	—	—	C	×
B1A79:11*4	—	Rear fog light circuit malfunction	—	—	C	×
B1A79:13*4	—	Rear fog light circuit malfunction	—	—	C	×
B1C79:13	—	Washer motor circuit malfunction	—	—	C	×
B1D00:11	—	Headlight LO (LH) circuit malfunction	—	—	C	×
B1D00:13	—	Headlight LO (LH) circuit malfunction	—	—	C	×
B1D01:11	—	Headlight LO (RH) circuit malfunction	—	—	C	×
B1D01:13	—	Headlight LO (RH) circuit malfunction	—	—	C	×
B1D02:11	—	Headlight HI (LH) circuit malfunction	—	—	C	×
B1D02:13	—	Headlight HI (LH) circuit malfunction	—	—	C	×
B1D03:11	—	Headlight HI (RH) circuit malfunction	—	—	C	×
B1D03:13	—	Headlight HI (RH) circuit malfunction	—	—	C	×
B1D39:11	—	Rear wiper motor circuit malfunction	—	—	C	×
B1D39:13	—	Rear wiper motor circuit malfunction	—	—	C	×
P0604:49	—	Electrical supply unit (ESU) internal malfunction	×	—	C	×
P0605:49	—	Electrical supply unit (ESU) internal malfunction	×	—	C	×

DTC No.	Warning/ indicator light	Description	Fail-safe	Drive cycle	Self test type*1	Memory function
U0140:00	—	Communication error with BCM	×	—	C	×
U0322:09	—	Error signal received from BCM	×	—	C	×
U0322:82	—	Error signal received from BCM	×	—	C	×
U0422:86	—	Error signal received from BCM	×	—	C	×
U2010:11	—	Illumination output circuit malfunction	—	—	C	×
U3000:09	—	Electrical supply unit (ESU) internal malfunction	—	—	C	×
U3000:44	—	Electrical supply unit (ESU) internal malfunction	×	—	C	×
U3000:47	—	Electrical supply unit (ESU) internal malfunction	—	—	C	×
U3000:49	—	Electrical supply unit (ESU) internal malfunction	—	—	C	×
U3003:16	—	Electrical supply unit (ESU) power supply voltage low input	—	—	C	×
U3003:17	—	Electrical supply unit (ESU) power supply voltage high input	—	—	C	×
U3006:16	—	Electrical supply unit (ESU) power supply voltage low input	—	—	C	×

*1 : C: CMDTC self test, D: ODDTC self test

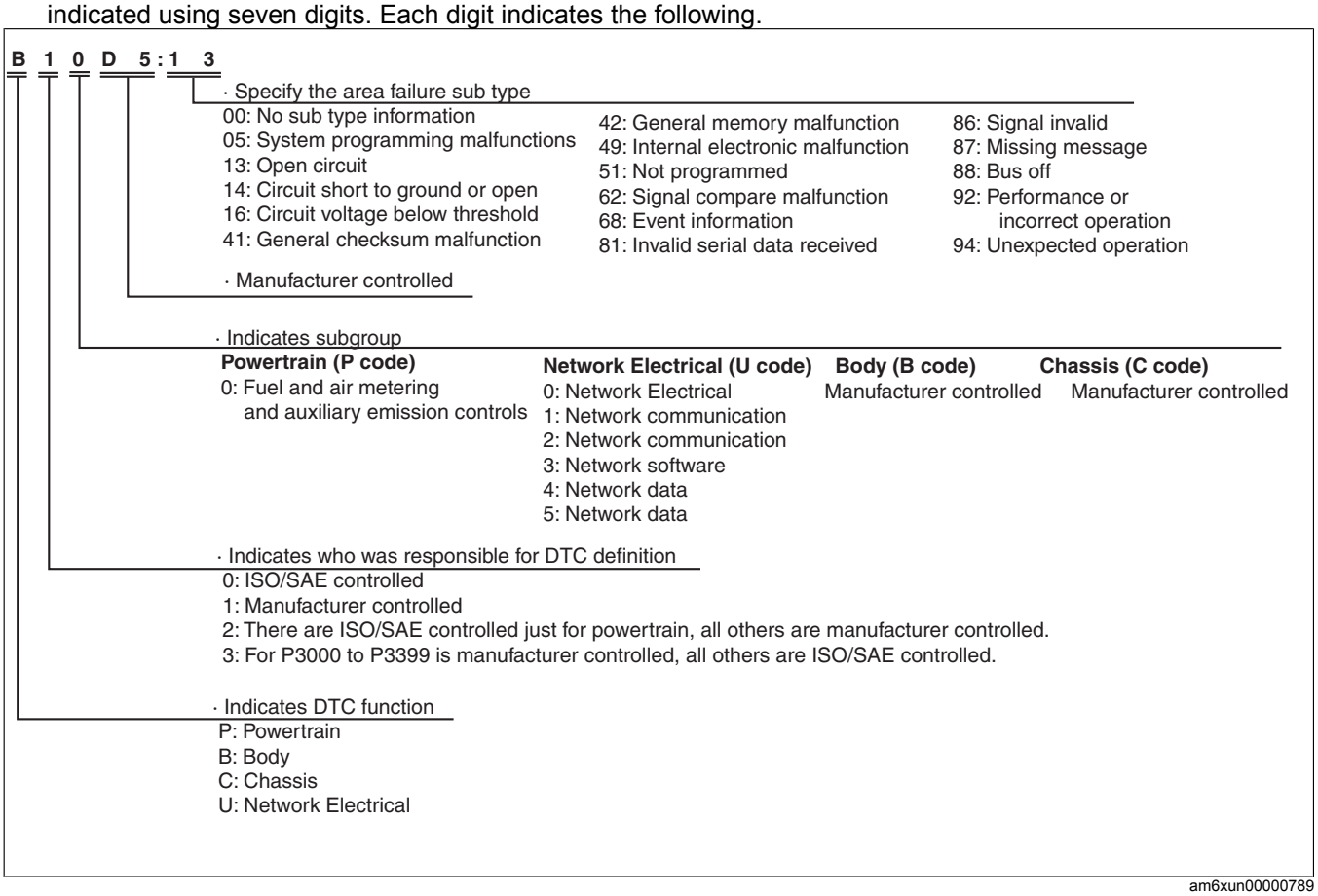
*2 : With collision detection unlock function

*3 : With running light system

*4 : With rear fog light system

DTC 7-digit code definition

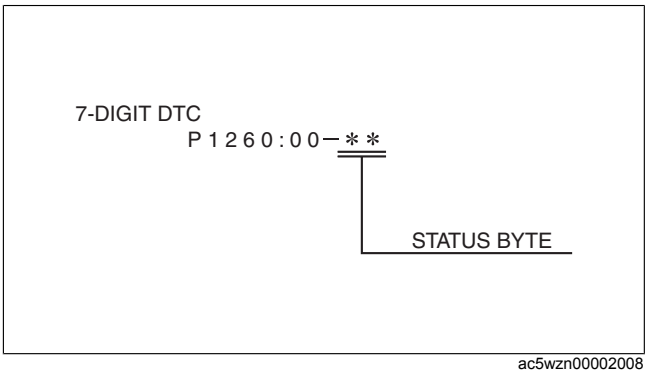
- When related systems or components have failed, the CM stores the DTC of the malfunctioning part in the CM memory, and allows for the retrieval of the store data using scanning tool when necessary. The DTCs are indicated using seven digits. Each digit indicates the following.



am6xun00000789

Status byte for DTC

- The two digits (two digits after hyphen (-)) after the 7-digit DTC.
- A code which indicates the pending code, current/ past malfunction status, or warning illumination status.
- Can be read by performing a CMDTC self-test using the Mazda Modular Diagnostic System (M-MDS).
- For details on the status byte, refer to the explanation on the Mazda Modular Diagnostic System (M-MDS) when reading the DTC.



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Detection condition for the applicable DTC

DTC No.	Description	Detection condition
B1098:11	Parking light (LH) or taillight (RH) circuit malfunction	• The electrical supply unit (ESU) detects a short to ground in the parking light (LH) or taillight (RH) circuit.
B1099:11	Parking light (RH), taillight (LH), or license plate light (LH)/(RH) circuit malfunction	• The electrical supply unit (ESU) detects a short to ground in the parking light (RH), taillight (LH), or license plate light (LH)/(RH) circuit.
B111E:12	**4SD** • Trunk compartment light circuit malfunction **5HB** • Cargo compartment light circuit malfunction	**4SD** • Electrical supply unit (ESU) detects a short to power supply in the trunk compartment light circuit. **5HB** • Electrical supply unit (ESU) detects a short to power supply in the cargo compartment light circuit.
B1142:13	Ignition switch status mismatch	• The electrical supply unit (ESU) detects the following mismatches. — Ignition switch status determined from power supply condition in BCM received from BCM via LIN communication — Ignition switch status determined from power supply input condition of electrical supply unit (ESU) power supply circuit (IG1)
B1207:09*2	Error signal received from SAS control module	• The electrical supply unit (ESU) detected the following conditions for **2 s or more** with the ignition switched ON (engine off or on). — A signal is received which cannot normally determine the collision conditions from SAS control module even though the collision detection unlock function presence signal received from the BCM determines that the function is present.
B1207:86*2	Error signal received from SAS control module	• The electrical supply unit (ESU) detected the following conditions for **2 s or more** with the ignition switched ON (engine off or on). — Error signal is received from SAS control module even though the collision detection unlock function presence signal received from the BCM determines that the function is present.
B124A:11*3	Running light circuit malfunction	• Electrical supply unit (ESU) detects a short to ground in the running light circuit with the ignition switched ON (engine off or on).
B1277:11	Back-up light circuit malfunction	• Electrical supply unit (ESU) detects a short to ground in the back-up light circuit with the ignition switched ON (engine off or on).
B1313:12	Front map light circuit malfunction	• Electrical supply unit (ESU) detects a short to power supply in the front map light circuit.
B143A:00*2	Collision detection unlock function operation	• Electrical supply unit (ESU) operates the collision detection unlock function.

DTC No.	Description	Detection condition
B1A79:11[*4]	Rear fog light circuit malfunction	• Electrical supply unit (ESU) detects a short to ground in the rear fog light circuit with the ignition switched ON (engine off or on).
B1A79:13[*4]	Rear fog light circuit malfunction	• Electrical supply unit (ESU) detects an open circuit in the rear fog light circuit with the ignition switched ON (engine off or on).
B1C79:13	Washer motor circuit malfunction	• Electrical supply unit (ESU) detects an open circuit in the washer motor circuit with the ignition switched ON (engine off or on).
B1D00:11	Headlight LO (LH) circuit malfunction	• Electrical supply unit (ESU) detects a short to ground in the headlight LO (LH) circuit.
B1D00:13	Headlight LO (LH) circuit malfunction	• Electrical supply unit (ESU) detects an open circuit in the headlight LO (LH) circuit.
B1D01:11	Headlight LO (RH) circuit malfunction	• Electrical supply unit (ESU) detects a short to ground in the headlight LO (RH) circuit.
B1D01:13	Headlight LO (RH) circuit malfunction	• Electrical supply unit (ESU) detects an open circuit in the headlight LO (RH) circuit.
B1D02:11	Headlight HI (LH) circuit malfunction	• Electrical supply unit (ESU) detects a short to ground in the headlight HI (LH) circuit.
B1D02:13	Headlight HI (LH) circuit malfunction	• Electrical supply unit (ESU) detects an open circuit in the headlight HI (LH) circuit.
B1D03:11	Headlight HI (RH) circuit malfunction	• Electrical supply unit (ESU) detects a short to ground in the Headlight HI (RH) circuit.
B1D03:13	Headlight HI (RH) circuit malfunction	• Electrical supply unit (ESU) detects an open circuit in the headlight HI (RH) circuit.
B1D39:11	Rear wiper motor circuit malfunction	• Electrical supply unit (ESU) detects a short to ground in the rear wiper motor circuit with the ignition switched ON (engine off or on).
B1D39:13	Rear wiper motor circuit malfunction	• Electrical supply unit (ESU) detects an open circuit in the rear wiper motor circuit with the ignition switched ON (engine off or on).
P0604:49	Electrical supply unit (ESU) internal malfunction	• RAM malfunction in electrical supply unit (ESU) detected.
P0605:49	Electrical supply unit (ESU) internal malfunction	• ROM malfunction in electrical supply unit (ESU) detected.
U0140:00	Communication error with BCM	• Electrical supply unit (ESU) could not receive the CAN signal from the BCM for 2 s or more with the ignition switched ON (engine off or on).
U0322:09	Error signal received from BCM	• Electrical supply unit (ESU) receives the error signals from the BCM for **2 s or more** with the ignition switched ON (engine off or on).
U0322:82	Error signal received from BCM	• Electrical supply unit (ESU) receives the error signals from the BCM for **2 s or more** with the ignition switched ON (engine off or on).
U0422:86	Error signal received from BCM	• Any of the following conditions is met with the ignition switched ON (engine off or on): — The electrical supply unit (ESU) received a headlight LO on request signal error signal from the BCM for **2 s or more**. — The electrical supply unit (ESU) received a headlight HI on request signal error signal from the BCM for **2 s or more**. — The electrical supply unit (ESU) detected a mismatch in the following signal combinations received from the BCM for **2 s or more**. • Headlight variation information signal • Headlight LO on request signal — The electrical supply unit (ESU) detected a mismatch in the following signal combinations received from the BCM for **2 s or more**. • Headlight variation information signal • Headlight HI on request signal

DTC No.	Description	Detection condition
U2010:11	Illumination output circuit malfunction	• Electrical supply unit (ESU) detects a short to ground in the illumination output circuit.
U3000:09	Electrical supply unit (ESU) internal malfunction	• Malfunction in the electrical supply unit (ESU) internal circuit (self-protection circuit) is detected.
U3000:44	Electrical supply unit (ESU) internal malfunction	• The electrical supply unit (ESU) detected a NVRAM malfunction in the electrical supply unit (ESU).
U3000:47	Electrical supply unit (ESU) internal malfunction	• The electrical supply unit (ESU) detected a watchdog malfunction in the electrical supply unit (ESU).
U3000:49	Electrical supply unit (ESU) internal malfunction	• RAM/ROM/ALU/TIMER malfunction in the electrical supply unit (ESU) is detected.
U3003:16	Electrical supply unit (ESU) power supply voltage low input	• Electrical supply unit (ESU) power supply circuit (terminal 1H, 2P, or 2R) of **less than 8 V** is detected for **10 s or more**.
U3003:17	Electrical supply unit (ESU) power supply voltage high input	• Electrical supply unit (ESU) power supply circuit (terminal 1H, 2P, or 2R) of **16 V or more** is detected for **5 s or more**.
U3006:16	Electrical supply unit (ESU) power supply voltage low input	• Electrical supply unit (ESU) power supply circuit voltage (terminal 2B) of **4 V or less** is detected for **5 s or more**.